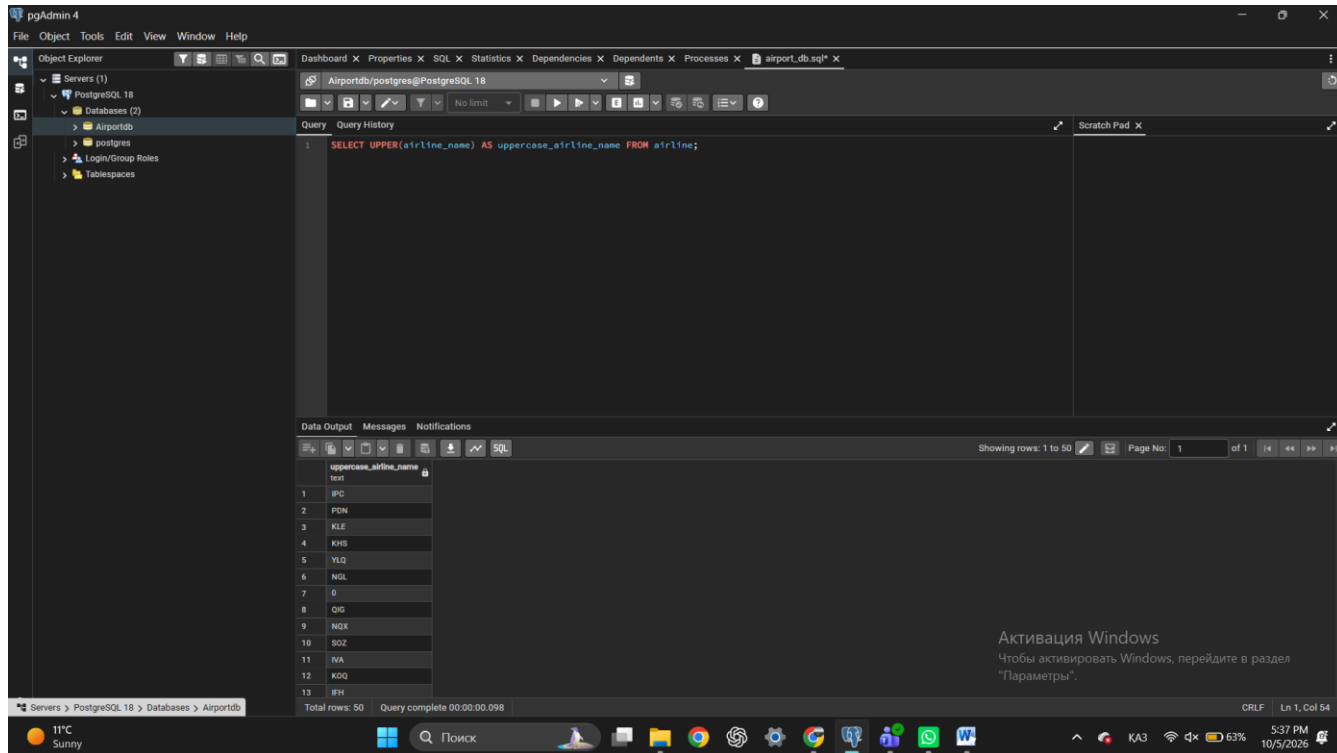


Laboratory work 4

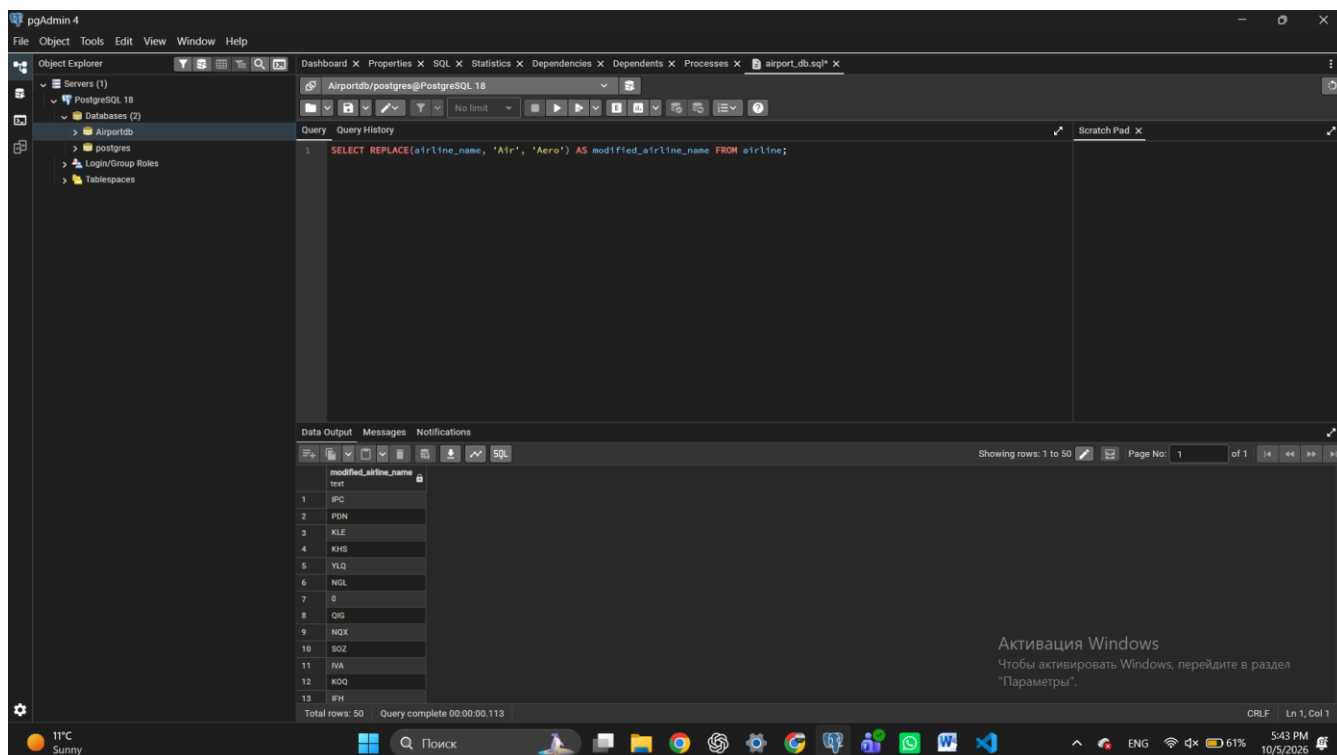
Darkhanova Mekhriniso

Tasks:

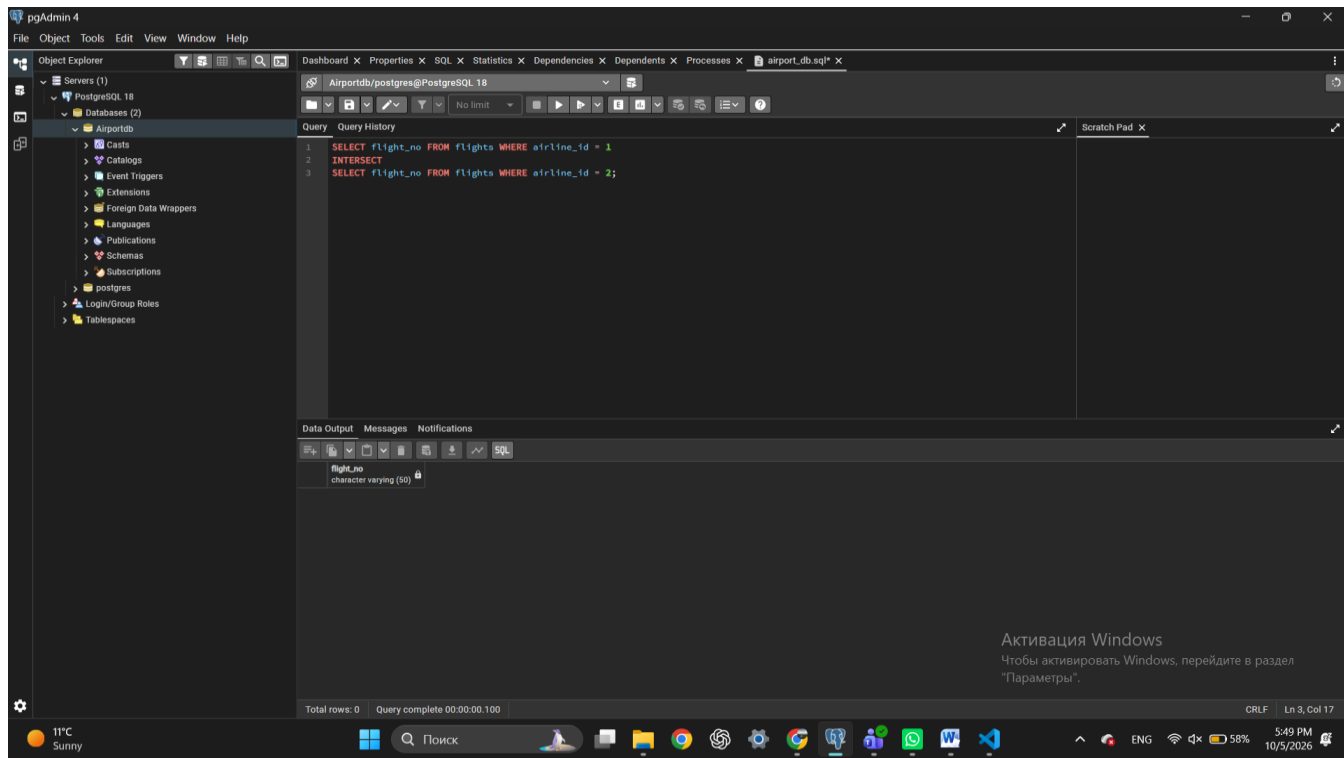
1. Retrieve all airline names in uppercase.



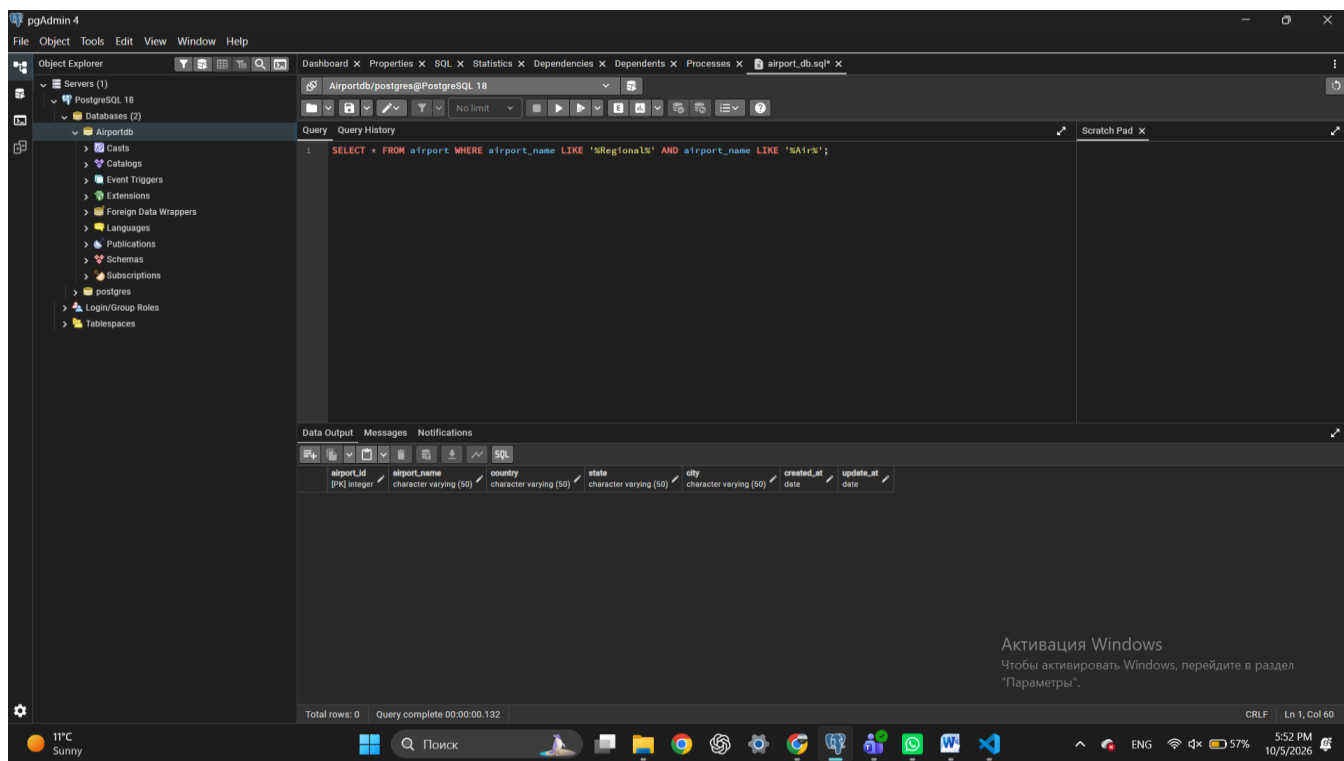
2. Replace any occurrence of the word "Air" in airline names with "Aero".



3. Find all flight numbers that coordinates with both airline 1 and airline 2.



- Retrieve airports that contain the word "Reginal" and "Air" in their names.



- Retrieve passenger names and format their birth dates as 'Month DD, YYYY'.

pgAdmin 4 interface showing a query executed on the `airportdb/postgres@PostgreSQL 18` database. The query is:

```
SELECT
  first_name,
  last_name,
  TO_CHAR(date_of_birth, 'FMMonth DD, YYYY') AS formatted_birth_date
FROM passengers;
```

The results are displayed in a table with 13 rows and 3 columns:

	first_name	last_name	formatted_birth_date
1	Hilde	Imis	January 03, 2000
2	Arvy	Sparsholt	June 09, 1974
3	Reinold	Pockocke	June 07, 1982
4	Con	Borrel	October 17, 1986
5	Wayne	Banga	April 22, 1996
6	Tildy	Shackelford	April 15, 2004
7	Byrle	Gram	July 07, 1985
8	Howard	Igo	March 29, 1984
9	Alphonso	Philippos	August 30, 1975
10	Fairfax	Needham	July 08, 1981
11	Vande	Prime	May 17, 1977
12	Tema	Wimmer	January 02, 1977
13	Domestic	Blaazczak	December 31, 2001

Total rows: 200. Query complete 00:00:00.176.

6. Find flight numbers that have been delayed based on the actual arrival time.

pgAdmin 4 interface showing a query executed on the `airportdb/postgres@PostgreSQL 18` database. The query is:

```
SELECT flight_no FROM flights WHERE actual_arrival > scheduled_arrival;
```

The results are displayed in a table with 13 rows and 1 column:

flight_no	
1	US-GT
2	US-NM
3	RU-KR
4	US-AZ
5	IN-OR
6	CA-NL
7	BR-PE
8	TH-32
9	US-MB
10	GB-ENG
11	UZ-SU
12	US-OR
13	KP-10

Total rows: 509. Query complete 00:00:00.103.

Successfully run. Total query runtime: 103 msec. 509 rows affected.

7. Find flights that arrived late according to their actual arrival time compared to the scheduled arrival time.

pgAdmin 4

Object Explorer: Servers (1) > PostgreSQL 18 > Databases (2) > Airportdb

Query: `SELECT * FROM flights WHERE actual_arrival > scheduled_arrival;`

Data Output:

flight_id	flight_no	scheduled_departure	scheduled_arrival	departure_airport_id	arrival_airport_id	departing_gate	arriving_gate	airline_id	status	actual_departure	actual_arrival
1	US-CT	2024-01-22	2023-09-08	12	15	9	649	39	Check-in open	2023-10-30	2023-11-07
2	US-NM	2023-07-21	2023-09-17	13	16	4216	90	34	Boarding	2024-02-09	2024-01-23
3	RU-KR	2024-01-02	2023-03-18	3	19	151	66	33	Check-in open	2023-10-10	2023-04-07
4	US-AZ	2023-07-29	2023-04-08	13	6	5045	1985	12	Delayed	2023-07-10	2023-08-01
5	IN-GR	2023-05-18	2023-09-19	6	9	659	621	13	Delayed	2023-06-17	2023-12-03
6	CA-NL	2023-03-22	2023-06-04	12	13	40	2592	27	Boarding	2023-04-28	2023-11-17
7	BR-PE	2024-01-16	2023-06-02	13	20	1891	5429	1	Delayed	2023-08-19	2023-11-09
8	TH-32	2024-02-05	2023-03-30	8	19	478	[null]	17	Boarding	2023-04-21	2023-07-21
9	US-MS	2024-02-06	2024-01-22	10	1	346	56	38	Delayed	2023-10-30	2024-03-06
10	GB-ENG	2023-11-03	2023-10-11	2	2	400	85	85	Delayed	2024-02-05	2024-01-04
11	UZ-SU	2023-05-13	2023-05-24	11	2	1027	190	206	Delayed	2024-02-24	2023-12-23
12	US-SU	2024-03-10	2023-07-14	6	14	214	285	2	Delayed	2023-06-20	2023-10-21

Total rows: 509 Query complete 00:00:00.155

Successfully run. Total query runtime: 155 msec. 509 rows affected.

8. Show airlines from any of: ('France','Portugal','Poland') created between '2023-11-01' and '2024-03-31'.

pgAdmin 4

Object Explorer: Servers (1) > PostgreSQL 18 > Databases (2) > Airportdb

Query: `SELECT * FROM airline WHERE airline_country IN ('France', 'Portugal', 'Poland') AND created_at BETWEEN '2023-11-01' AND '2024-03-31';`

Data Output:

airline_id	airline_code	airline_name	airline_country	created_at	update_at
4	OOKB	KHS	France	2023-11-18	2024-02-18
13	OIFE	IFH	Portugal	2024-01-09	2023-11-09
15	SVKZ	KKG	Poland	2023-12-24	2023-10-30
24	ZSRG	RUG	Portugal	2024-03-11	2023-08-24
28	RKPD	JDG	Poland	2023-11-03	2023-12-06
30	CYST	YST	Poland	2023-11-24	2023-10-07
43	ATKM	KMA	France	2023-11-21	2023-05-30

Total rows: 7 Query complete 00:00:00.190

9. Find top3 overweighted baggage with more than 25kg.

pgAdmin 4

Object Explorer

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Query

```
SELECT * FROM baggage WHERE weight_in_kg > 25 ORDER BY weight_in_kg DESC LIMIT 3;
```

Data Output

baggage_id [PK] integer	weight_in_kg numeric (4,2)	created_date date	update_date date	booking_id integer	
1	321	49.80	2023-07-07	2023-11-10	349
2	61	49.57	2023-04-21	2023-07-31	464
3	102	49.44	2024-01-24	2023-11-01	53

Total rows: 3 Query complete 00:00:00.175

Активация Windows
Чтобы активировать Windows, перейдите в раздел "Параметры".
Successfully run. Total query runtime: 175 msec. 3 rows affected.

10. Find the youngest passengers' full name.

pgAdmin 4

Object Explorer

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Query

```
SELECT first_name, last_name FROM passengers ORDER BY date_of_birth DESC LIMIT 1;
```

Data Output

first_name character varying (50)	last_name character varying (50)
Bernie	Michal

Total rows: 1 Query complete 00:00:00.164

Активация Windows
Чтобы активировать Windows, перейдите в раздел "Параметры".

11. Find the cheapest booking price on each booking platform.

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

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Dashboard Properties SQL Statistics Dependencies Dependents Processes airport_db.sql

Query

```
1 SELECT
2   booking_platform,
3   MIN(price) AS cheapest_price
4 FROM booking GROUP BY booking_platform;
```

Query History

Scratch Pad

Data Output Messages Notifications

Showing rows: 1 to 494 Page No: 1 of 1

	booking_platform	cheapest_price
	character varying (50)	numeric
1	Schmitt Quigley	4966.68
2	Rohan, Hamill and Bosco	7118.84
3	Kutch Russel	2113.46
4	Veum Corwin	2183.14
5	Wolff, Wuckert and Satterfield	4911.89
6	Reichel, Deckow and Quitzon	8389.34
7	Murazik-Gulgowski	950.80
8	Cole, Corwin and Leannon	8735.70
9	Franecki, Dickens and Walsh	283.78
10	McLaughlin, Spies and Abernathy	7480.57
11	Leffler-Thompson	2559.77
12	Raynor Inc	410.47
13	Waelchi Inc	3694.97

Total rows: 494 Query complete 00:00:00.129

11°C Sunny

Активация Windows
Чтобы активировать Windows, перейдите в раздел "Параметры".

12. Return airlines whose airline_code contains a digit.

pgAdmin 4

File Object Tools Edit View Window Help

Object Explorer

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Dashboard Properties SQL Statistics Dependencies Dependents Processes airport_db.sql

Query

```
1 SELECT * FROM airline WHERE airline_code ~ '[0-9]';
```

Query History

Scratch Pad

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

airline_id	airline_code	airline_name	airline_country	created_at	update_at
[PK] integer	character varying (50)	character varying (50)	character varying (50)	date	date
1	2AK6	HGZ	Philippines	2024-03-13	2023-10-20

Total rows: 1 Query complete 00:00:00.137

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Активация Windows
Чтобы активировать Windows, перейдите в раздел "Параметры".

13. List the top5 most recently created airlines.

pgAdmin 4

Object Explorer: Servers (1) PostgreSQL 18 Databases (2) Airportdb

Query: `SELECT * FROM airline ORDER BY created_at DESC LIMIT 5;`

Data Output:

	airline_id [PK] integer	airline_code character varying (50)	airline_name character varying (50)	airline_country character varying (50)	created_at date	update_at date
1	9	KNQX	NOX	Venezuela	2024-03-16	2023-09-16
2	40	LTBD	CH	Argentina	2024-03-15	2024-01-29
3	22	2AK6	HQZ	Philippines	2024-03-13	2023-10-20
4	24	ZSRG	RUG	Portugal	2024-03-11	2023-08-24
5	45	HCMV	BUO	Sweden	2024-02-18	2023-12-10

Total rows: 5 Query complete 00:00:00.200

Активация Windows
Чтобы активировать Windows, перейдите в раздел "Параметры".
Successfully run. Total query runtime: 200 msec. 5 rows affected.

14. Return all rows where booking_id is between 200 and 300 inclusive and check_result <> 'Checked'.

pgAdmin 4

Object Explorer: Servers (1) PostgreSQL 18 Databases (2) Airportdb

Query: `SELECT * FROM baggage_check WHERE booking_id BETWEEN 200 AND 300 AND check_result <> 'Checked';`

Data Output:

	baggage_check_id [PK] integer	check_result character varying (50)	created_at date	update_at date	booking_id integer	passenger_id integer
1	7	Not checked	2023-05-04	2023-04-29	226	44
2	12	Not checked	2023-10-02	2023-06-22	288	133
3	22	Not checked	2023-06-14	2023-11-10	246	181
4	23	Not checked	2023-05-29	2023-06-02	284	22
5	24	Not checked	2024-02-17	2023-10-03	267	68
6	25	Not checked	2024-01-30	2023-06-10	237	140
7	61	Not checked	2023-06-12	2023-11-05	279	195
8	73	Not checked	2023-12-02	2023-05-28	274	187
9	75	Not checked	2023-11-03	2023-12-03	300	130
10	83	Not checked	2023-10-04	2023-04-26	259	63
11	91	Not checked	2023-05-21	2023-09-03	234	76
12	92	Not checked	2024-01-26	2023-04-07	260	154
13	107	Not checked	2024-03-06	2023-05-01	218	188

Total rows: 56 Query complete 00:00:00.160

Активация Windows
Чтобы активировать Windows, перейдите в раздел "Параметры".
Successfully run. Total query runtime: 160 msec. 56 rows affected.

15. Baggage checks where update_at is in the same month as created_at but occurs earlier than created_at.

pgAdmin 4

File Object Tools Edit View Window Help

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Dashboard X Properties X SQL X Statistics X Dependencies X Dependents X Processes X airport_db.sql* X

Airportdb/postgres@PostgreSQL 18

Query Query History

```
1 SELECT * FROM baggage_check WHERE EXTRACT(MONTH FROM update_at) - EXTRACT(MONTH FROM created_at)
2 AND update_at < created_at;
```

Data Output Messages Notifications

	baggage_check_id (PK) integer	check_result character varying (50)	created_at date	update_at date	booking_id integer	passenger_id integer
1	34	Checked	2023-04-25	2023-04-08	28	57
2	86	Checked	2023-12-25	2023-12-07	312	22
3	104	Checked	2024-03-17	2024-03-14	41	83
4	109	Not checked	2023-08-30	2023-08-09	469	91
5	119	Not checked	2023-12-02	2023-12-01	135	55
6	148	Checked	2024-03-17	2024-03-16	86	5
7	219	Checked	2023-04-17	2023-04-14	233	200
8	248	Not checked	2023-06-22	2023-06-09	449	60
9	251	Not checked	2023-06-19	2023-06-16	260	26
10	299	Checked	2023-04-26	2023-04-08	385	191
11	304	Not checked	2023-07-29	2023-07-22	210	10
12	314	Not checked	2024-03-07	2023-03-28	254	57
13	396	Not checked	2023-07-13	2023-07-01	37	144

Total rows: 21 Query complete 00:00:00.146

Showing rows: 1 to 21 Page No: 1 of 1

Активация Windows
Чтобы активировать Windows, перейдите в раздел "Параметры".

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Поиск

6:15 PM 10/5/2026